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Whole-genome sequence and phenotypic characteristics of *Bacillus cereus* strain IM8 isolated from the Upper Fremont Glacier in Wyoming

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ABSTRACT *Bacillus cereus* strain IM8 isolated from an ice core of the Upper Fremont Glacier in Wyoming showed 97.52% average nucleotide identity with *B. cereus sensu stricto*, a lineage typically not associated with psychrotolerance. Phenotypic characterization highlighted high cytotoxicity toward human intestinal epithelial Caco-2 cells and resistance to ampicillin.

KEYWORDS *Bacillus cereus*, glacier

Glacial environments serve as natural cryogenic libraries where microorganisms can remain preserved for multimillennial periods, providing opportunities to study them (1–3). The *Bacillus cereus* group comprises closely related, spore-forming gram-positive species of environmental, medical, and agricultural importance (4, 5). Here, we report the genome and phenotypic characterization of *B. cereus* IM8 isolated from the Fremont Glacier in Wyoming.

A 401.2 g ice core from the Upper Fremont Glacier, Wyoming (43°07′37″ N, 109°36′54″ W) was collected in 1991 at a 28 m depth using a sterile solar-powered drill (6) (Fig. 1) and stored at −20°C. On 8 April 2025, the core was melted aseptically, and 25 mL aliquots were filtered through 0.2 µm Whatman membranes (Millipore Sigma). Filters were plated on tryptic soy agar (TSA, BD) and incubated at 22°C for 48 h. Twenty-two colonies were isolated and preserved in 20% glycerol at −80°C. 16S rRNA genes were amplified as previously described (7) and Sanger sequenced at the University of Chicago Sequencing Center. National Center for Biotechnology Information (NCBI) BLAST analysis suggested that isolate IM8 was *B. cereus*, having 99.26% nucleotide identity, 100% query coverage, and an *E* value of 0 to *B. cereus* strain NPK1_1_39.

DNA extraction, WGS, and genome assembly were performed by SeqCenter (Pittsburgh, PA). An overnight culture streaked on TSA (BD) at 23°C was sent to SeqCenter. DNA extraction was performed with the ZymoBIOMICS DNA Miniprep Kit (Zymo Research), and DNA was quantified with Qubit 4 (Thermo Fisher Scientific). For Nanopore sequencing, libraries were prepared using 60 µL of DNA (40 ng/µL) with the Ligation Sequencing Kit (SQK-LSK114) and NEBNext Companion Module (E7180L) following the manufacturer's instructions. No additional DNA fragmentation or size selection was performed. Nanopore sequencing was performed on a GridION using R10.4.1 flow cells, and Dorado (Version 0.5.3, super-accurate and 5mC/5hmC models) was used for basecalling and demultiplexing. For Illumina sequencing, libraries were prepared using 30 µL of DNA (10 ng/µL) with the Illumina DNA Prep Kit following the manufacturer's instructions using custom 10 bp unique dual indices (UDI) and a target insert size of 280 bp. Illumina sequencing was performed on an Illumina NovaSeq X Plus sequencer, producing 2 × 151 bp paired-end reads. Demultiplexing, quality control, and adapter trimming were performed with bcl-convert1 (Version 4.2.4). Nanopore reads

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The authors declare no conflict of interest.

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